

Part I

SUMMARY of the Limit Laws

Summary

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0.1 The Limit Laws:

Continuity If a function f is **continuous at** a , then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Composition Limit Law If a function f is **continuous at** $b = \lim_{x \rightarrow a} g(x)$, then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

For the limit laws below, assume $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ both exist. The following properties hold.

Sum Limit Law

$$\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

Difference Limit Law

$$\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

Product Limit Law

$$\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \left(\lim_{x \rightarrow a} f(x)\right) \cdot \left(\lim_{x \rightarrow a} g(x)\right)$$

Quotient Limit Law

$$\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \text{ provided } \lim_{x \rightarrow a} g(x) \neq 0$$

Limit laws also hold for one-sided limits.

Part II

Recitation Questions

Problem 1

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Problem 1 The following argument shows

$$\lim_{x \rightarrow 3} \frac{5x^3 - 4\sqrt{x}}{\sqrt{x^5 - 87}} = \frac{135 - 4\sqrt{3}}{\sqrt{156}}.$$

State which limit law is used to justify each step. (A particular step may have more than one limit law as a justification.)

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{5x^3 - 4\sqrt{x}}{\sqrt{x^5 - 87}} &= \frac{\lim_{x \rightarrow 3} (5x^3 - 4\sqrt{x})}{\lim_{x \rightarrow 3} \sqrt{x^5 - 87}} \\ &= \frac{\left(\lim_{x \rightarrow 3} 5 \right) \left(\lim_{x \rightarrow 3} (x^3) \right) - \left(\lim_{x \rightarrow 3} 4 \right) \left(\lim_{x \rightarrow 3} \sqrt{x} \right)}{\sqrt{\lim_{x \rightarrow 3} (x^5 - 87)}} \\ &= \frac{5 \lim_{x \rightarrow 3} (x^3) - 4 \lim_{x \rightarrow 3} \sqrt{x}}{\sqrt{\lim_{x \rightarrow 3} (x^5 - 87)}} \\ &= \frac{5(\lim_{x \rightarrow 3} x)^3 - 4\sqrt{3}}{\sqrt{\lim_{x \rightarrow 3} (x^5) - \lim_{x \rightarrow 3} (87)}} \\ &= \frac{5(3)^3 - 4\sqrt{3}}{\sqrt{3^5 - 87}} \\ &= \frac{135 - 4\sqrt{3}}{\sqrt{156}} \end{aligned}$$

Problem 2

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Problem 2

Problem 2 Find the limit and justify your answer.

(a) $\lim_{x \rightarrow 0} |x|$

(b) $\lim_{x \rightarrow 2} \ln \left(\sin(x - 2) + e^x \sin \left(\frac{\pi x}{4} \right) \right)$

(c) $\lim_{x \rightarrow 1} \frac{3 + 2 \cos \left(\frac{\pi x}{3} \right)}{4x - 2x^3}$

Problem 3

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Problem 3

Problem 3 Suppose $f(x) = \begin{cases} x^2 - ax & \text{if } x < 3 \\ a2^x + 7 + a & \text{if } x > 3 \end{cases}$

Find a so that $\lim_{x \rightarrow 3} f(x)$ exists.

Problem 4

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Problem 4

Problem 4 Determine the value of $\lim_{x \rightarrow 0} \left(x^2 \cos \left(\frac{1}{x} \right) \right)$. **EXPLAIN.**

Problem 5

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Problem 5

Problem 5 For all x near 0, the inequalities $1 - \frac{x^2}{6} \leq \frac{\sin(x)}{x} \leq 1$ are true.

Use these inequalities to find $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$. **EXPLAIN.**

Problem 6

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Problem 6

Problem 6 Two functions, h and g , are given

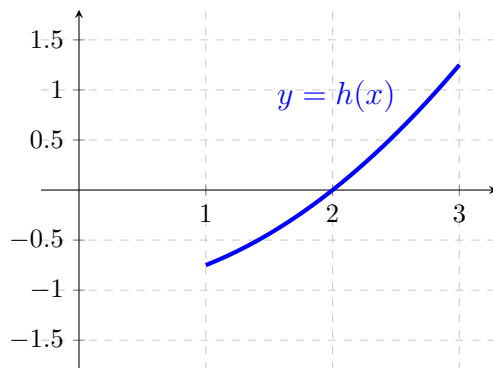
$$h(x) = \frac{x^2 - 4}{4}, \quad 1 < x < 3$$

$$g(x) = x - 2, \quad 1 < x < 3$$

The graph of the function h is given in the figure below. Let f be a function defined on the interval $(1, 3)$ that satisfies the following inequalities

$$g(x) \leq f(x) \leq h(x), \quad 1 < x < 3$$

- (a) In the figure below, sketch and label the graph of g and a possible graph of f . (All three functions have a common domain $(1, 3)$.)



- (b) Evaluate the limit, or state that it does not exist. Justify your answer.

- (i) $\lim_{x \rightarrow 2} f(x)$
- (ii) $\lim_{x \rightarrow 2} \frac{f(x) + 2}{x - 1}$
- (iii) $\lim_{x \rightarrow 2} g(1 + e^{f(x)})$